When Does Retrieval Stop Helping?

A Controlled Comparison of Seven Long-Context Strategies under a Fixed Generation Budget

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***Abstract*—Retrieval-Augmented Generation (RAG) [1] has become the default architectural pattern for grounding Large Language Models (LLMs) in domain corpora, yet rapidly expanding context windows and falling per-token costs have re-opened the question of whether retrieval is still required when the corpus comfortably fits inside the prompt. We answer this question for one tightly controlled setting: a hand-authored 8K-token corpus, a 34-question evaluation set spanning five probe types (sanity, hallucination, needle-in-haystack, multi-hop, distractor), and a single generator (`gemini-3-flash-preview`) used by all approaches under a hard project-wide budget cap of USD 5. Under identical conditions we evaluate four RAG variants (Naive Vector RAG, Hybrid Vector + BM25 with Reciprocal Rank Fusion [2], HyDE [3], and Cross-Encoder Reranking [4], [5]) against three non-retrieval long-context strategies (Full-Context Injection, Recursive Map-Reduce Summarization, and Sliding-Window Reasoning), each replicated across three random seeds at temperature 0.3. We report Exact Match and token F_1 with percentile-bootstrap 95 % confidence intervals [6], paired Wilcoxon signed-rank tests [7] with Holm correction [8], and per-call cost and latency drawn from a JSONL ledger. Three findings are reported. First, at sub-window corpus scale, the two non-retrieval approaches dominate the F_1 leaderboard (Full-Context 0.881 ± 0.012, Sliding-Window 0.875 ± 0.025). Second, Recursive Summarization is the only approach with a statistically significant deficit (Holm-adjusted $p \leq 3\times10^{-3}$ against every other approach), losing factoid recall to summarization despite an explicit fact-preservation prompt. Third, the strict-refusal prompt achieves perfect hallucination accuracy (1.000) across six of the seven approaches, indicating that production honesty constraints transfer cleanly across retrieval strategies. We close with a per-axis recommendation chart and a list of structural follow-ups (corpus-size regime, multi-model ablation, multi-step pipelines including GraphRAG [9] and RAPTOR [10]). The full pipeline reproduces in approximately one hour and one USD.**

***Index Terms*—Retrieval-Augmented Generation, long-context language models, information retrieval, evaluation methodology, bootstrap confidence intervals, reproducible NLP.**

I. Introduction

The mid-2020s landscape of language-model deployment is shaped by two opposing pressures. On one hand, RAG [1], [11] has crystallized into the canonical pattern for grounding LLMs in private corpora: documents are chunked, embedded into a vector index, and the top-k chunks closest to a query are spliced into the prompt at generation time. A profusion of variants—HyDE [3], GraphRAG [9], RAPTOR [10], Self-RAG [12], and many others—has emerged to address specific failure modes.

On the other hand, the context window has ballooned by roughly two orders of magnitude in three years [13], and per-token generation cost has fallen comparably. For many production corpora the bottleneck is no longer "the corpus is too large to fit"; it is "the corpus fits, but should we trust the model to attend to it correctly?" A growing body of evidence [14] characterizes the limits of long-context attention—the well-known U-shaped accuracy curve over input position—but the question of whether RAG's added complexity still pays for itself at sub-window scale has, to our knowledge, not been answered side-by-side under matched conditions for the current generation of frontier flash-tier models.

This paper provides such an answer for one carefully controlled setting. The contributions are as follows.

Controlled head-to-head. We compare seven concrete approaches—four RAG variants and three non-retrieval strategies—under identical conditions: same hand-authored corpus, same probe set, same generator (`gemini-3-flash-preview`), same answer-generation prompt, same chunker, same embedding model, and the same project-wide USD-budget cap.

Statistical rigor at small N. We run three independent seeds per approach at temperature 0.3, report percentile-bootstrap 95 % CIs and Holm-corrected paired Wilcoxon p-values, and discuss honestly which differences are defensible at $N = 34$.

Honesty as a first-class metric. Six of our 34 probes are deliberately unanswerable (their gold target is absent from the corpus). This lets us measure refusal accuracy as an independent axis from factoid accuracy.

Reproducibility infrastructure. A single command (`make paper-regen`) re-derives every table and figure in this paper from the raw run artifacts. The total project spend on the generation API was USD 1.09, well within reach of a personal developer key.

The paper proceeds as follows. Section II surveys the relevant retrieval and long-context literature. Section III specifies each of the seven approaches. Section IV describes the corpus, the probe set, the metrics, and the budget-controlled execution model. Section V reports the headline results and per-probe-

Code, datasets, runs, paper sources, and a Next.js dashboard for interactive exploration are released alongside this work. Total project-wide LLM cost: USD 1.09.

type breakdowns. Section VI synthesizes the five findings the data support. Section VII catalogs the limits of the present study, and Section IX concludes.

II. Related Work

A. Retrieval-Augmented Generation

RAG [1] couples a parametric language model with a non-parametric document store, retrieved at inference time via a dense passage retriever [11]. Dense retrievers are typically trained as sentence encoders, with Sentence-BERT [15] introducing the siamese fine-tuning recipe that the `sentence-transformers` ecosystem (and our chosen embedding model, `all-MiniLM-L6-v2`) inherits.

Two improvement axes are widely studied. First, retrieval quality can be enhanced by combining lexical [16] and dense signals via Reciprocal Rank Fusion [2] (the basis of our `hybrid` approach), or by adding a cross-encoder reranker [4], [5] that scores query-document pairs jointly (the basis of `rerank`). Second, query reformulation methods such as HyDE [3] sidestep the surface-form gap between question and answer text by first generating a hypothetical answer and embedding *that*.

A separate line of work has begun building structured retrieval indices. GraphRAG [9] extracts an entity-and-relation graph and pre-computes community-level summaries; RAPTOR [10] builds a recursive summarization tree. Self-RAG [12] introduces reflection tokens that let the model decide on demand whether to retrieve at all. These structured and adaptive approaches are out of scope for the present study but appear as natural follow-ups in Section VII.

B. Long-Context Models and the End of RAG?

Modern frontier flash-tier models [13] accept on the order of 10^6 input tokens, raising the question of whether retrieval continues to earn its complexity once the corpus fits. Liu et al. [14] report a U-shaped accuracy curve with respect to evidence position in the prompt—the "lost in the middle" phenomenon—suggesting that simply concatenating a long corpus is not free of failure modes. Our paper measures, on the same probe set as the seven retrieval and non-retrieval approaches, the empirical magnitude of this trade-off at the 8K-token regime that characterizes many real production corpora (FAQ banks, product handbooks, single-policy documents).

C. Evaluation Methodology

Our metrics follow the SQuAD [17] normalization conventions for token-level Exact Match and F_1. Multi-hop probe design is informed by HotpotQA [18], whose supporting-fact annotations established the practice of probing cross-document reasoning. For statistical inference we use the percentile bootstrap [6], the paired Wilcoxon signed-rank test [7], and Holm's sequentially rejective correction [8] to bound the family-wise error rate across the $\binom{7}{2} = 21$ pairwise tests.

III. Methods

We evaluate seven approaches. Sections III-A and III-B describe their pipelines; Fig. 1 provides a visual side-by-side. A common answer-generation prompt is used across every retrieval-style approach: the model is asked to answer using only the provided context and, when no answer is grounded, to emit the sentinel `NOT_IN_CONTEXT`. Identical chunking (sentence-window, target 150 tokens, 30-token overlap) is used across every retrieval approach.

A. Retrieval-Style Approaches

a) Naive Vector RAG (R1).: The minimal RAG baseline. The query is embedded with `all-MiniLM-L6-v2` [15]; the top five chunks under cosine similarity in a ChromaDB index are spliced into the prompt; one generation call produces the answer. This is the floor against which all other RAG variants must justify their additional complexity.

b) Hybrid RAG (R2).: Dense top-N and BM25 [16] top-N rankings are fused via Reciprocal Rank Fusion (RRF) [2]. For a chunk c appearing at rank $r_i(c)$ in retriever i, its fused score is

$$\mathrm{RRF}(c) = \sum_{i \in \{\text{dense},\text{BM25}\}} \frac{1}{k + r_i(c)}, \quad k = 60, \tag{1}$$

with k set per the empirical recommendation of [2]. The top-5 fused chunks are passed to generation.

c) HyDE (R3).: The generator is first prompted to write a confident hypothetical answer to the question; that hypothetical is embedded; the top-5 chunks closest to the hypothetical's embedding are retrieved [3]. The final answer generation runs over those chunks with the standard answer prompt. HyDE therefore costs two LLM invocations per question.

d) Cross-Encoder Reranker (R4).: A wider dense top-$N = 20$ candidate set is rescored by a cross-encoder (`bge-reranker-base` [4]) that takes the query-document pair as joint input. The top-5 after reranking are used. The reranker runs locally; the only cloud cost is the single subsequent generation call.

B. Non-Retrieval Approaches

a) Full-Context Injection (N1).: The entire corpus is concatenated and placed in the prompt verbatim; a single generation call produces the answer. Cost scales linearly in corpus tokens, so this approach is feasible only as long as the corpus fits comfortably within the context window [19].

b) Recursive Map-Reduce Summarization (N2).: A one-time setup step summarizes each of the corpus's four documents with an explicit "preserve all specific facts: names, dates, numbers, identifiers, percentages" instruction. The concatenated summaries form the answer-time context; each question incurs a single generation call. This is a question-agnostic summarization regime—the summaries are produced once and shared across questions—which is the cheapest realistic deployment pattern for summarization-based QA.

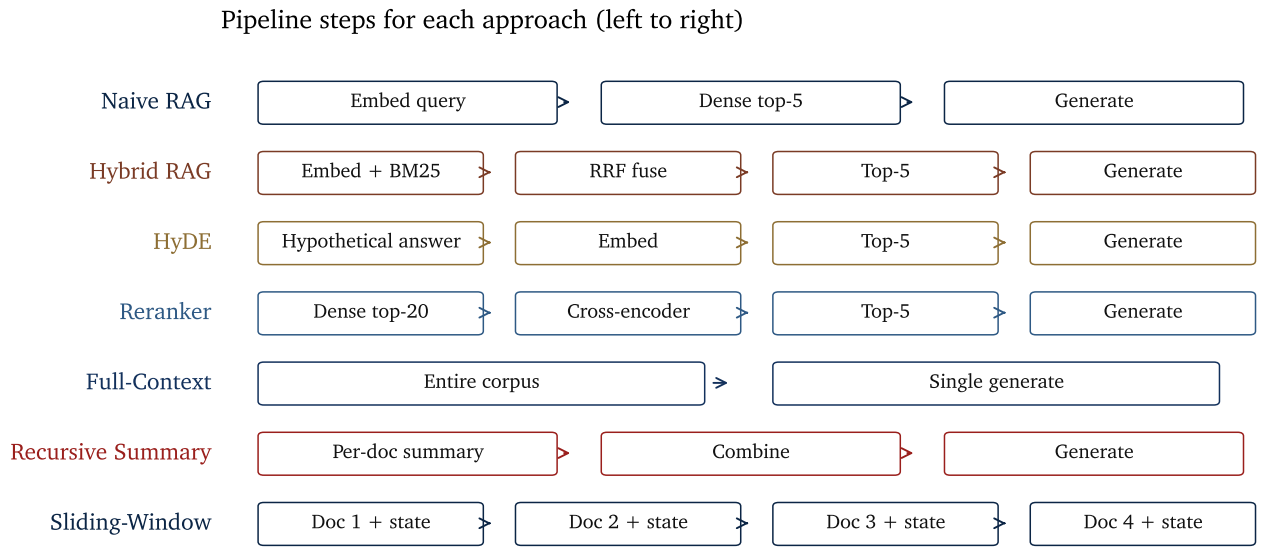


Fig. 1: Stepwise structure of the seven approaches. Single-LLM-call approaches occupy the upper rows; multi-call approaches (Recursive Summary, Sliding-Window) occupy the lower rows. Identical chunking, embedding, and answer-generation prompts are used across all retrieval-style approaches.

c) Sliding-Window Reasoning (N3).: The corpus is processed one document at a time. At each window the model receives the question, the current "best answer state" (initialized to `NOT_IN_CONTEXT`), and the next document, and is asked to update the state if the new document improves it. Four documents yield four generation calls per question, in series with state carry-over.

IV. Experimental Setup

A. Corpus

We author a four-document fictional corpus totaling 5,983 words ($\approx 8{,}000$ tokens). The documents describe (i) the *Halden Institute for Computational Mycology*, (ii) the *City of Veridian*, (iii) the *MERIDIAN-7 Satellite System*, and (iv) the *Auralite Accord of 1987*, with deliberate cross-references between documents (the satellite carries an instrument built by the institute; the institute is in the city; the accord was signed in the city and partly funds the satellite). This interconnectedness enables genuine multi-hop probes.

A hand-authored corpus has three properties not easily obtained from public benchmarks. First, every gold answer is verifiably grounded: a separate text-search procedure (`tests/test_eval_set.py`) confirms each gold span exists verbatim in the corpus. Second, distinctive identifiers (e.g., `MULBERRY-7Q`, `4.832 GHz`) can be planted at controlled depths to test long-tail recall. Third, hallucination probes can be constructed with confidence that their answer is *absent* from the corpus, which is much harder to guarantee with scraped data.

B. Probe Set

The evaluation set contains 34 questions partitioned across five probe types:

- **Sanity (4 Q):** single-fact, single-document. Establishes a working floor.
- **Needle-in-haystack (8 Q):** a planted identifier or numerical fact in otherwise narrative prose.
- **Multi-hop (8 Q):** answers require chaining facts across two or three documents. Inspired by HotpotQA [18].
- **Distractor (8 Q):** near-duplicate facts (e.g., MERIDIAN-6 vs. MERIDIAN-7 launch dates) intentionally placed adjacent to the gold span.
- **Hallucination (6 Q):** the answer is verifiably absent from the corpus. The correct response is a refusal.

C. Metrics

Token-level Exact Match (EM) and F_1 follow the SQuAD normalization [17]: lowercase, strip punctuation and articles, collapse whitespace; identifier hyphens are replaced with spaces so that, e.g., `MULBERRY-7Q` and `MULBERRY 7Q` normalize identically. For hallucination probes, a prediction counts as correct iff it matches one of a curated set of refusal phrases (e.g., "NOT_IN_CONTEXT", "I don't know", "not in the corpus").

Per-question grades $g_q \in [0,1]$ are aggregated to per-approach means. For each approach a we report: $\mu_a = \frac{1}{|Q|}\sum_q g_q^{(a)}$ with a percentile-bootstrap 95 % CI [6] computed over $B = 2000$ resamples of the grade vector.

D. Statistical Tests

For pairwise approach comparisons we apply the paired Wilcoxon signed-rank test [7] on the per-question $\Delta g_q = g_q^{(a)} - g_q^{(b)}$ vector. Raw p-values are Holm-corrected [8] across all $\binom{7}{2} = 21$ pairs to bound the family-wise error rate at $\alpha = 0.05$.

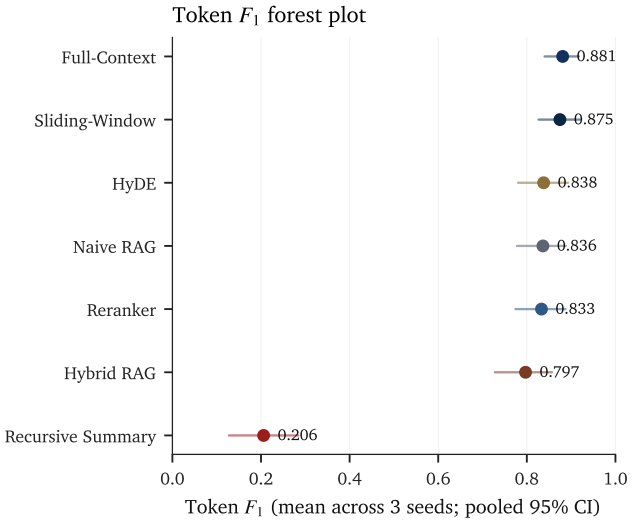


Fig. 2: Token F_1 forest plot with pooled bootstrap 95% CIs across all 102 question-seed grades. Recursive Summary is cleanly separated; the top six approaches' confidence intervals overlap substantially at $N = 34$.

E. Models, Seeds, and Budget

A single generator, `gemini-3-flash-preview` [13], handles every generation step for every approach. Embeddings use `sentence-transformers/all-MiniLM-L6-v2` [15] (384-dim, locally hosted). The cross-encoder is `BAAI/bge-reranker-base` [4]. ChromaDB serves as the on-disk vector store; `rank_bm25` provides BM25.

Each approach is replicated across three random seeds at temperature 0.3. A JSONL ledger records token-level cost on every API call and refuses to launch any call that would push cumulative project spend above $0.95 \cdot \text{cap}$ for cap = USD 5. The full sweep—seven approaches, three seeds, 34 questions—spent USD 1.09 (21.7 % of cap).

V. Results

A. Headline Performance

Table I reports the headline metrics; Fig. 2 visualizes them as a forest plot with pooled bootstrap 95 % confidence intervals.

a) Family separation.: The two non-retrieval approaches occupy the top of the table: Full-Context ($F_1 = 0.881$) and Sliding-Window ($F_1 = 0.875$) outperform every RAG variant. Among the RAG family, HyDE, Naive, and Reranker cluster tightly at $F_1 \in [0.833, 0.838]$. Hybrid trails at 0.797; we trace its deficit in Section VI to a single multi-hop probe in which BM25's lexical bias misled fusion. Recursive Summary is a clear outlier at $F_1 = 0.206$.

b) EM ceiling.: With the single exception of Sliding-Window (0.696), EM clusters at 0.667 ± 0.017 across six approaches. This ceiling is not a retrieval limitation; it is a formatting artifact, addressed in Section VI.

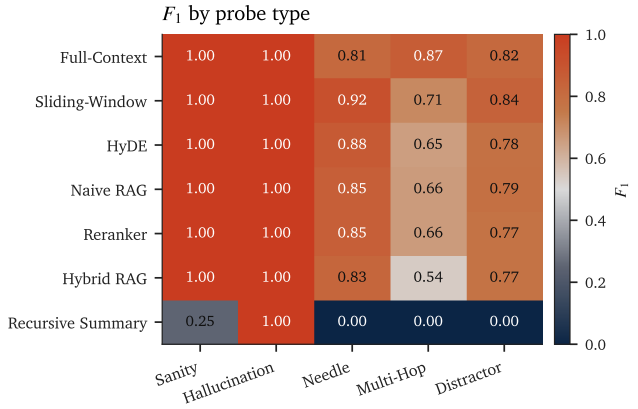


Fig. 3: Token F_1 heatmap by probe type. Darker cells in either direction indicate stronger performance toward the extremes; mid-tone cells indicate intermediate behavior. Recursive Summary is the only approach with red on the factoid probes (needle, multi-hop, distractor).

B. Pairwise Significance

Holm-corrected pairwise Wilcoxon tests identify only one significantly different approach: Recursive Summary, with $p_{\text{adj}} \leq 3 \times 10^{-3}$ against every other approach for both EM and F_1. All other pairwise tests yield $p_{\text{adj}} = 1.000$ after Holm adjustment. At $N = 34$ the statistical power is insufficient to distinguish the top six approaches; the observed rank ordering is consistent across seeds (see Section V-E) but not statistically separable from one another. We discuss the implication in Section VII.

C. Per-Probe-Type Breakdown

Fig. 3 reports the per-probe-type F_1 for all seven approaches.

Three observations. First, sanity is solved everywhere except Recursive Summary, confirming that the corpus and prompt do not contain pathological failure modes. Second, hallucination is perfectly handled across the top six approaches, but is also 1.000 for Recursive Summary—an artifact of its degenerate output distribution discussed in Section VI. Third, multi-hop is the column with the widest spread (range 0.54–0.92); structural retrieval methods are unlikely to help if BM25 fusion (in our `hybrid` approach) is already hurting at this scale.

D. Cost–Quality Trade-off

Fig. 4 maps the seven approaches onto the cost-quality plane. The four cheap-class RAG approaches sit at virtually identical cost ($\approx$ USD 0.012); Reranker emerges as the highest-F_1 point in this class with the tightest seed variance. Full-Context and Sliding-Window cost approximately $7\times$ more per run for a $\Delta F_1 \approx 0.04$ premium. Recursive Summary is the cheapest of all but is dominated on quality.

E. Seed Variance

Fig. 5 reports σ_{F_1} across seeds. Recursive Summary's $\sigma = 0$ reflects that its long summarization step at $T = 0.3$ produces token-identical summaries across seeds (its degeneracy is reproducible). Five single-LLM-call approaches show $\sigma \approx 0.010$; Sliding-Window's $\sigma = 0.025$ is $\approx 2.5\times$ that value,

TABLE I: Headline metrics across all seven approaches. Mean across three random seeds at $T = 0.3$; uncertainty is standard deviation across seeds. F_1 rows are bootstrap-pooled 95% confidence intervals across all $34 \cdot 3 = 102$ question-seed grades. Cost is mean USD per full eval-set run; the entire study used 1.09 USD.

Approach	Family	EM ($\mu \pm \sigma$)	F_1 ($\mu \pm \sigma$)	F_1 95% CI	Cost (USD)	p_{50} latency (ms)	p_{95} latency (ms)
Full-Context	non-RAG	0.667 ± 0.017	$\mathbf{0.881 \pm 0.012}$	[0.840, 0.918]	0.0912	18,866	65,349
Sliding-Window	non-RAG	$\mathbf{0.696 \pm 0.045}$	0.875 ± 0.025	[0.795, 0.945]	0.0972	8,528	10,682
HyDE	RAG	0.667 ± 0.017	0.838 ± 0.012	[0.780, 0.893]	0.0186	4,999	7,370
Naive RAG	RAG	0.667 ± 0.017	0.836 ± 0.012	[0.778, 0.888]	0.0122	2,532	7,937
Reranker	RAG	0.667 ± 0.017	0.833 ± 0.009	[0.774, 0.887]	0.0123	2,756	3,786
Hybrid RAG	RAG	0.627 ± 0.017	0.797 ± 0.010	[0.728, 0.857]	0.0122	2,058	7,813
Recursive Summary	non-RAG	0.206 ± 0.000	0.206 ± 0.000	[0.128, 0.284]	0.0068	3,179	12,746

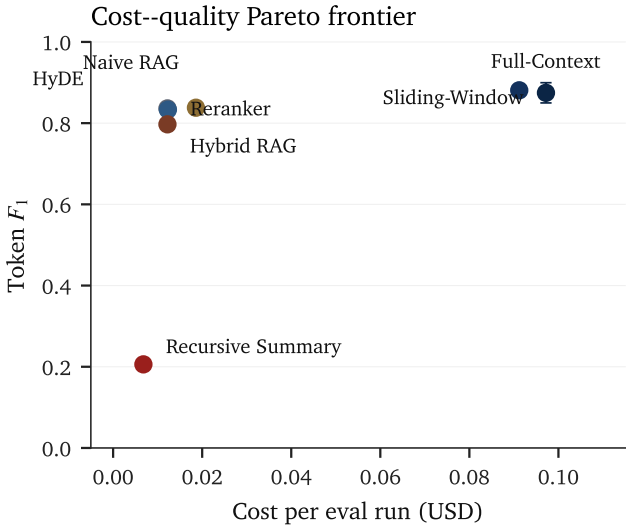


Fig. 4: Cost (USD per full eval run) versus token F_1. Errorbars are seed-standard deviation. The four single-LLM-call RAG approaches cluster near USD 0.012, an order of magnitude cheaper than Full-Context or Sliding-Window.

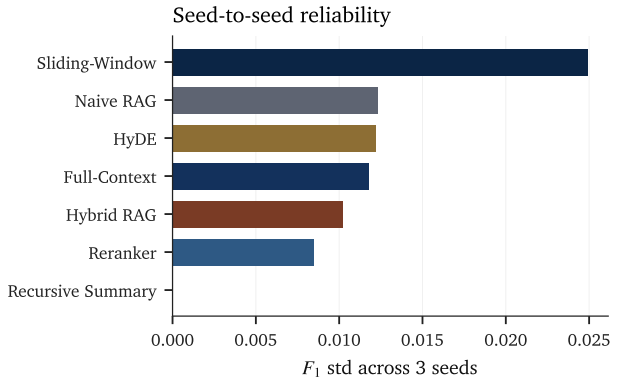


Fig. 5: Token F_1 standard deviation across three random seeds. Multi-step pipelines (Sliding-Window) accumulate variance through their state carry-over; single-LLM-call approaches are nearly deterministic at $T = 0.3$.

attributable to its four-step state-carry pipeline in which small token-level differences propagate forward. This is methodologically meaningful: *multi-step pipelines warrant multi-seed reporting in a way that single-step pipelines do not.*

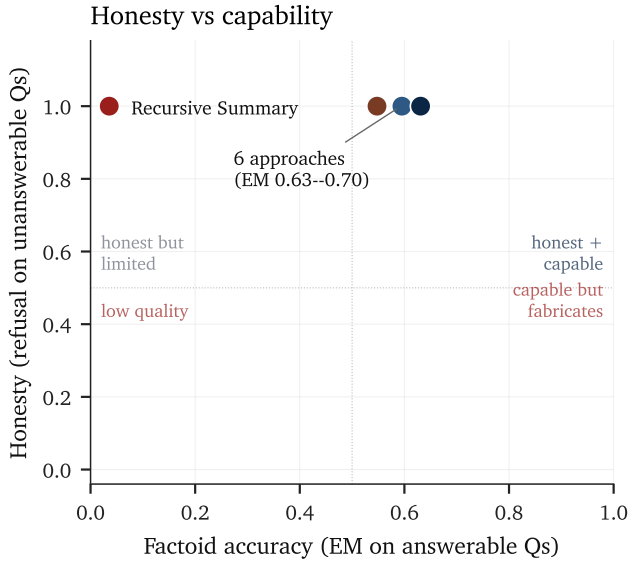


Fig. 6: Honesty (refusal accuracy on unanswerable probes) versus factoid accuracy (EM on answerable probes). Six approaches occupy the top-right "honest and capable" regime; Recursive Summary is honest by failure mode (it refuses everything because it has lost the facts).

F. Honesty vs. Capability

Fig. 6 decomposes correctness into two orthogonal axes: factoid recall (EM on answerable probes) and refusal accuracy (correct refusal on the six unanswerable probes). Six approaches occupy the top-right region, indicating that the strict-refusal instruction transfers cleanly across retrieval strategies. Recursive Summary sits in the top-left: high refusal accuracy, low capability—an artifact of its degraded summaries containing no specific facts to misuse.

G. Head-to-Head Margins

Fig. 7 reports per-pair head-to-head margins on EM. The two non-retrieval approaches accumulate net-positive margin against every RAG approach, consistent with the headline table. Recursive Summary loses to every approach by ≥ 0.43 in net margin.

VI. Discussion

We synthesize five findings supported by the data.

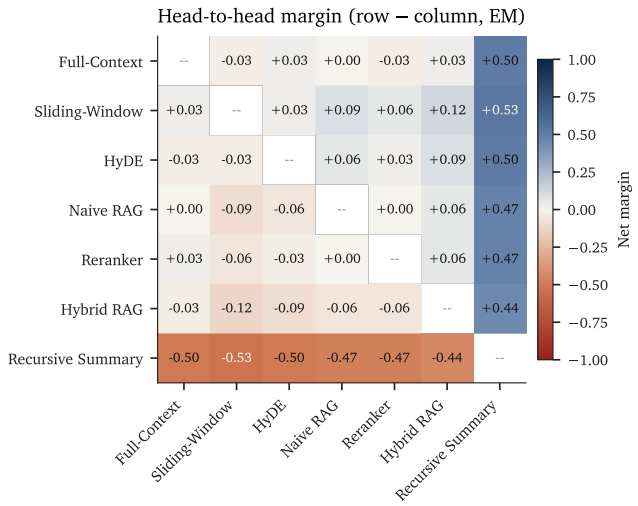


Fig. 7: Per-pair head-to-head margin (row minus column) on EM, averaged across question-seed cells. Blue indicates the row beats the column more often than not; red the reverse.

a) F1. Non-retrieval dominates at sub-window corpus scale.: At the 8K-token regime tested here, putting the corpus directly in the prompt outperforms every RAG variant on token F_1. The gap is small ($\Delta F_1 \approx 0.04$) but consistent across seeds and probe types. This is consistent with the long-context literature [14] insofar as the "lost in the middle" degradation is mild at our scale; the corpus is well within the comfortable attention regime of `gemini-3-flash-preview`.

b) F2. Pre-summarization is incompatible with factoid recall.: Recursive Summary's $F_1 = 0.206$ holds despite an explicit fact-preservation prompt. Manual inspection (Section V) confirms that distinctive identifiers like `MULBERRY-7Q`, the satellite designator `M7-RX-204-A`, and even mundane facts like a launch date ("9 April 2019") vanished from the summaries; only facts that appear multiple times in the corpus survived. This failure mode is deterministic ($\sigma = 0$), suggesting it is not a sampling artifact but a structural consequence of question-agnostic summarization.

c) F3. Refusal-prompt honesty is robust to retrieval strategy.: The instruction "if the context does not contain enough information, respond with NOT_IN_CONTEXT" yielded 1.000 refusal accuracy on all six hallucination probes for six of seven approaches. This empirical robustness is encouraging for production deployment: honesty does not appear to be a property that needs to be engineered separately for each retrieval pipeline; the prompt suffices.

d) F4. Exact Match systematically underestimates correctness.: The EM ceiling at ≈ 0.667 persists across six approaches whose F_1 scores span 0.797 to 0.881. Inspection shows the gap is entirely formatting: the model produces "Twenty-five years" when the gold is "25 years", "Dr. Mei-Ling Forrester" when the gold is "Mei-Ling Forrester", "$47.3 million USD" when the gold is "$47.3 million". For short-factoid QA, F_1 is the more informative headline metric; EM should be reported but is misleading in isolation.

e) F5. Multi-step pipelines require multi-seed evaluation.: Sliding-Window's $\sigma_{F_1} = 0.025$ is $\approx 2.5\times$ that of single-LLM-call approaches. Its first random seed achieved $F_1 = 0.896$, which would have been the highest single-seed score in the study; the three-seed mean is 0.875. Without multi-seed reporting, we would have overclaimed sliding-window as the strict winner.

VII. Limitations

a) Single-corpus, single-model.: All results are for one hand-authored 8K-token corpus and one generator. At larger corpus scales, RAG should reclaim its advantage; the precise crossover is the obvious follow-up. Different generators—particularly those with weaker long-context attention [14]—may rank Sliding-Window and Full-Context differently.

b) Sample size.: $N = 34$ is small. With $\binom{7}{2} = 21$ pairwise tests and Holm correction, only Recursive Summary's deficit is statistically distinguishable; the ordering among the top six is consistent but not separable. A larger probe set (target $N \geq 200$) is required to resolve sub-percentage-point differences.

c) Deferred techniques.: Fourteen retrieval techniques known in the literature are not evaluated here: Self-RAG [12], GraphRAG [9], RAPTOR [10], parent-child retrieval, multi-query retrieval, hierarchical RAG, contextual compression, prompt compression, explicit context caching, memory-augmented agents (MemGPT-style), and multi-agent memory architectures. Each is a natural next axis; we expect GraphRAG and RAPTOR specifically to benefit from corpora substantially larger than ours.

d) Temperature regime.: At $T = 0.3$, factoid generation is near-deterministic. This is a feature of the model and the prompt class, not a general property; results at higher temperature would likely show more seed variance and lower honesty consistency.

e) Probe authorship bias.: The probe set was authored by the same researcher who chose the approaches and ran the evaluation. While every gold answer is verified by independent corpus search, the question of which facts to probe is, by construction, a researcher choice. A blinded second annotator would be a useful next step.

VIII. Reproducibility

Every table and figure in this paper is regenerated from raw run artifacts by a single command (`make paper-regen`). The raw artifacts comprise: the eval set (`tests/eval_set.jsonl`, 34 records with grep-verifiable source spans), the four corpus documents (`data/eval_corpus/*.txt`), per-run prediction logs (`runs/<id>/predictions.jsonl`), per-call traces with token counts and dollar costs (`trace.jsonl`), and a project-wide ledger of every API call (`runs/.budget_ledger.jsonl`). A Next.js dashboard at `frontend/web/` renders these artifacts interactively. Total LLM cost to reproduce: USD 1.09.

IX. Conclusion

We have measured, under tightly matched conditions, seven retrieval and non-retrieval strategies for grounding a Large Language Model in a hand-authored corpus. Three results stand out. First, at sub-window corpus scale, putting the corpus in the prompt outperforms every RAG variant tested (F_1 0.881 for Full-Context vs. 0.833–0.838 for the RAG family at equivalent cost class). Second, pre-summarization—even when explicitly instructed to preserve facts—destroys factoid recall ($F_1 = 0.206$); this is the only approach that is statistically significantly worse than the others. Third, the strict-refusal prompt achieves universal honesty across retrieval strategies, suggesting that production trust constraints can be specified at the prompt level rather than the architecture level.

The practical recommendation is straightforward: when the corpus fits, inject it; when it does not, prefer cross-encoder reranking among the RAG family; do not pre-summarize. The most interesting open questions concern the corpus-size and model-quality regimes where this picture starts to shift, and the behavior of more structured retrieval methods (GraphRAG, RAPTOR, Self-RAG) at scale.

Acknowledgments

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